

UGB Solution

Aryan Goutam · UGB Subjective Portion · @MathVanguard

Topic: Trigonometry + Quadratic Equations • ISI Mathematics Pattern

Evaluation Protocol: Each problem carries **10 marks**. Marks are awarded for logical continuity and structural rigor. Providing final answers without full deduction yields **0 marks**. No negative penalties apply.

1 Problem 1 — Complex Quadratic with Modulus Constraint

Problem Statement

Let z_1, z_2 be complex numbers such that

$$z_1^2 - 4z_2 = 16 + 20i.$$

Suppose the roots α and β of

$$x^2 + z_1x + z_2 + m = 0$$

for some complex number m satisfy $|\alpha - \beta| = 2\sqrt{7}$.

- Determine the **maximum** value of $|m|$.
- Determine the **minimum** value of $|m|$.

By Vieta's formulas on the given equation, the sum and product of the roots are given by

$$\alpha + \beta = -z_1 \quad \text{and} \quad \alpha\beta = z_2 + m.$$

Expanding the algebraic expansion for the squared difference of the roots yields

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = z_1^2 - 4(z_2 + m) = (z_1^2 - 4z_2) - 4m.$$

Substituting the initial structural constraint $z_1^2 - 4z_2 = 16 + 20i$, we find

$$(\alpha - \beta)^2 = (16 + 20i) - 4m.$$

Applying the modulus boundary condition $|\alpha - \beta| = 2\sqrt{7}$, it follows that $|\alpha - \beta|^2 = 28$. Evaluating the magnitude on both sides gives

$$|(16 + 20i) - 4m| = 28 \implies 4|m - (4 + 5i)| = 28,$$

which simplifies directly to the locus equation

$$|m - (4 + 5i)| = 7.$$

Geometrically, m traces a circle of radius $r = 7$ centered at $c = 4 + 5i$ in the complex plane. The absolute distance of the center from the origin is

$$|c| = \sqrt{4^2 + 5^2} = \sqrt{41}.$$

By the triangle inequality, the range of $|m|$ is bounded by

$$||m| - |c|| \leq |m - c| = 7 \implies |c| - 7 \leq |m| \leq |c| + 7.$$

Since $\sqrt{41} < 7$, the lower bound evaluates algebraically to a negative value, implying the absolute spatial minimum is governed by the boundary constraint $|m| \geq 0$. Thus, the effective minimum distance reduces to $7 - \sqrt{41}$. Both extremes are collinear with the origin and the point $c = 4 + 5i$, matching locations where $m = (4 + 5i) \left(1 \pm \frac{7}{\sqrt{41}}\right)$.

$$(a) |m|_{\max} = \sqrt{41} + 7 \quad (b) |m|_{\min} = 7 - \sqrt{41}$$

2 Problem 2 — Floor Function System and Minimum Boundary

Problem Statement

Find all positive integers n such that 1991 is the minimum value of

$$f(k) = k^2 + \left\lfloor \frac{n}{k^2} \right\rfloor,$$

where k ranges over all positive integers.

The condition requires $f(k) \geq 1991$ for all $k \in \mathbb{Z}^+$, with strict equality holding for at least one choice of k . Isolating the floor function term gives

$$\left\lfloor \frac{n}{k^2} \right\rfloor \geq 1991 - k^2.$$

Because the right-hand side is strictly integer-valued, this inequality simplifies to the underlying rational form

$$\frac{n}{k^2} \geq 1991 - k^2 \implies n \geq k^2(1991 - k^2).$$

Consequently, the global validation threshold for n must satisfy

$$n \geq \max_{k \geq 1} k^2(1991 - k^2).$$

Let $g(x) = x(1991 - x)$ defined over $\mathbb{R}$. The vertex of this parabolic expression occurs at $x = \frac{1991}{2} = 995.5$. Since $x = k^2$ is constrained to the set of perfect squares, we test the bounding integers adjacent to the vertex location:

$$31^2 = 961 \quad \text{and} \quad 32^2 = 1024.$$

Evaluating $g(x)$ at these respective structural nodes yields

$$g(961) = 961(1991 - 961) = 961 \cdot 1030 = 989,830,$$

$$g(1024) = 1024(1991 - 1024) = 1024 \cdot 967 = 990,208.$$

Comparing these values shows the maximum is achieved when $k = 32$, which imposes the absolute lower bound

$$n \geq 990,208.$$

To ensure the global minimum of $f(k)$ matches 1991 exactly, we force the equality condition at the critical index $k = 32$:

$$1024 + \left\lfloor \frac{n}{1024} \right\rfloor = 1991 \implies \left\lfloor \frac{n}{1024} \right\rfloor = 967.$$

Unwrapping this floor inequality yields the standard system bounds

$$967 \leq \frac{n}{1024} < 968 \implies 967 \cdot 1024 \leq n < 968 \cdot 1024,$$

which computes to

$$990,208 \leq n < 991,232.$$

Restricting our solution set strictly to the integers, we establish the definitive range.

$$n \in \{990,208, 990,209, \dots, 991,231\}$$

$$\boxed{990,208 \leq n \leq 991,231}$$

Total solution count: $991,231 - 990,208 + 1 = 1024$ values.

3 Problem 3 — Bounded Quadratic Maximization

Problem Statement

For any real number $a < 0$, there exists a largest positive real number $\ell(a)$ such that

$$|ax^2 + 8x + 3| \leq 5$$

for all real x satisfying $0 \leq x \leq \ell(a)$.

- (a) Determine the **maximum** value of $\ell(a)$.
- (b) Determine a for which $\ell(a)$ is maximum.

Let $p(x) = ax^2 + 8x + 3$. For $x = 0$, $p(0) = 3$, meaning the tracking interval naturally satisfies the absolute constraint initially since $|3| \leq 5$. Splitting the system into dual linear inequalities gives

$$\begin{aligned} p(x) \leq 5 &\iff ax^2 + 8x - 2 \leq 0, \\ p(x) \geq -5 &\iff ax^2 + 8x + 8 \geq 0. \end{aligned}$$

Given $a < 0$, $p(x)$ describes a downward parabola whose axis of symmetry is located at $x^* = -\frac{4}{a} > 0$. The local maximum value at this point evaluates to

$$p(x^*) = 3 - \frac{16}{a}.$$

The peak profile reaches past the upper threshold boundary if $3 - \frac{16}{a} > 5$, which holds precisely when $a < -8$.

Analysis of Parameter Regimes

Regime A: $-8 < a < 0$

Here, the vertex coordinate exceeds the maximum tolerance line, meaning the upper threshold curve cuts the path first. The primary intersection locus for $ax^2 + 8x - 2 = 0$ features a positive discriminant $\Delta = 64 + 8a > 0$. The closest root restricting the interval domain is

$$x_1 = \frac{8 - \sqrt{64 + 8a}}{-2a}.$$

Substituting $w = \sqrt{64 + 8a} \in (0, 8)$ and reorganizing parameters yields

$$\ell_A(a) = \frac{8 - w}{-2a} = \frac{4}{8 + w}.$$

As $w \rightarrow 0^+$ (or equivalently, $a \rightarrow -8^+$), this limiting span approaches $\ell_A \rightarrow \frac{1}{2}$.

Regime B: $a \leq -8$

In this framework, the apex point remains below or exactly at the threshold

ceiling ($\Delta = 64 + 8a \leq 0$). Thus, the upper bound criteria is satisfied unconditionally across the domain. The length boundary is determined exclusively by the lower path limit intersection, where $ax^2 + 8x + 8 = 0$. The positive root is

$$\ell_B(a) = \frac{8 + \sqrt{64 - 32a}}{-2a}.$$

Setting $v = \sqrt{64 - 32a} \geq 8\sqrt{5}$, this simplifies to

$$\ell_B(a) = \frac{32}{v - 8}.$$

As a falls below -8 , the radical value v increases monotonically, causing $\ell_B(a)$ to contract. The maximum value occurs precisely at the transition interface $a = -8$, where

$$\ell_B(-8) = \frac{8 + \sqrt{64 + 256}}{16} = \frac{8 + 8\sqrt{5}}{16} = \frac{1 + \sqrt{5}}{2}.$$

Comparing both regimes, the global maximum interval length matches the golden ratio $\varphi = \frac{1+\sqrt{5}}{2}$, which is strictly achieved at the parameter setting $a = -8$.

$$(a) \ell_{\max} = \frac{1 + \sqrt{5}}{2} \quad (b) a = -8$$

4 Problem 4 — Parametric Trigonometric Field Constraints

Problem Statement

Determine the range of the real number θ such that for all $x \in [0, 1]$:

$$x^2 \cos \theta - x(1 - x) + (1 - x)^2 \sin \theta > 0.$$

Evaluating the boundaries of the domain interval gives necessary conditions for the parameters:

$$\text{At } x = 0 : (0) \cos \theta - 0 + (1) \sin \theta > 0 \implies \sin \theta > 0,$$

$$\text{At } x = 1 : (1) \cos \theta - 0 + 0 > 0 \implies \cos \theta > 0.$$

Combining these conditions locates θ firmly in the first quadrant, modulo complete rotations: $\theta \in (2k\pi, \frac{\pi}{2} + 2k\pi)$ where $k \in \mathbb{Z}$.

For the open interior domain $0 < x < 1$, we divide by $x^2 > 0$ and introduce the substitute variable $u = \frac{1-x}{x} > 0$:

$$\cos \theta - u + u^2 \sin \theta > 0 \quad \forall u > 0.$$

We define the quadratic model function $h(u) = u^2 \sin \theta - u + \cos \theta$. Because $\sin \theta > 0$, $h(u)$ represents a concave-up parabola. The minimum value occurs at the coordinate

$$u^* = \frac{1}{2 \sin \theta}.$$

Substituting this location back into the model expression yields the minimum value:

$$h(u^*) = \sin \theta \left(\frac{1}{4 \sin^2 \theta} \right) - \frac{1}{2 \sin \theta} + \cos \theta = \cos \theta - \frac{1}{4 \sin \theta}.$$

To verify the global positivity condition, this local minimum must satisfy $h(u^*) > 0$:

$$\cos \theta > \frac{1}{4 \sin \theta} \implies 4 \sin \theta \cos \theta > 1 \implies 2 \sin 2\theta > 1.$$

This simplifies to the condition $\sin 2\theta > \frac{1}{2}$. The continuous solution range for this functional inequality yields

$$2\theta \in \left(\frac{\pi}{6} + 2k\pi, \frac{5\pi}{6} + 2k\pi \right) \implies \theta \in \left(\frac{\pi}{12} + k\pi, \frac{5\pi}{12} + k\pi \right).$$

Intersecting this solution set with our original first-quadrant boundary rule requires k to be even ($k = 2n$). Odd values of k project into the third quadrant, where $\sin \theta < 0$, which violates the primary boundary criteria. Checking the endpoints shows that at $\theta = \frac{\pi}{12}$ or $\frac{5\pi}{12}$, $\sin 2\theta = \frac{1}{2}$, causing the strict inequality to fail. Thus, the boundary coordinates are omitted.

$$\theta \in \bigcup_{n \in \mathbb{Z}} \left(\frac{\pi}{12} + 2n\pi, \frac{5\pi}{12} + 2n\pi \right)$$

Summary of Results

Problem	Part	Answer
1	(a) $\max m $	$\sqrt{41} + 7$
1	(b) $\min m $	$7 - \sqrt{41}$
2	All valid n	$n \in \{990,208, \dots, 991,231\}$ (1024 values)
3	(a) $\max \ell(a)$	$\frac{1 + \sqrt{5}}{2}$ (golden ratio φ)
3	(b) Optimal a	$a = -8$
4	Range of θ	$\theta \in \left(\frac{\pi}{12} + 2n\pi, \frac{5\pi}{12} + 2n\pi \right), n \in \mathbb{Z}$

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